INCH-POUND

MIL-P-80279
27 November 1991

MILITARY SPECIFICATION

PUNCHING, SHEARING, NOTCHING, AND COPING MACHINE , METAL, HYDRAULICALLY DRIVEN, FOUR STATION TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hydraulically driven machines capable of performing punching, structural shape shearing, coping, and shearing operations on steel bars, structural shapes and plates of the dimensions and strengths specified herein.

1.2 Classification. The machine shall be of the following sizes. The size to be furnished shall be as specified (see 3.5 and 6.2). Size is determined by nominal punch capacity in tons and throat depth in inches.

Size	Nominal Punch Capacity (Tons)	Punch Throat Depth (Inches)
1	35	5
2	45	9
3	50	6
4	70	9
5	70	20
6	90	12
7	90	20
8	105	20
9	120	20
10	150	24

Beneficial comments (recommedations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Menphis, TN 38114-5213, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requiremnts for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurenmt of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 40, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

[Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited In the solicitation. Unless otherwise specified the Issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.5 - Iron Workers - Safety Requirements for the Construction, Care and Use. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

NATIONAL FLUID POWER ASSOCIATION, INC. (NFPA)

NFPA/JIC T2.24.1 - Hydraulic Fluid Power - Systems Standard for Industrial Machinery.

Application for copies should be addressed to the National Fluid Power Association Inc., 3333 N. Mayfair Road, Milwaukee, WI 53222-3291.

(Non Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein. the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design The machine shall be new (not a prototype) and one of the manufacturer s current production models capable of operations in accordance with the requirements herein. The machine shall be provided with not less than four separate and permanent metalworking stations. Unless otherwise specified (see 6.2), the machine shall accomplish not less than the four metalworking operations of punching, structural shape shearing, coping, and flat bar shearing. The machine shall be designed to operate using a non-flammable hydraulic fluid. The hydraulic unit shall be powered by an electric rotor and metalworking operations shall be accomplished by one or more hydraulic cylinders. The frame and base of the machine shall be an integral unit designed to rest and be mounted on a flat surface. Provisions suitable for hoisting the machine shall be provided. The base shall provide a means for securely bolting the machine to the floor or foundation. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6. 5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy In normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All operating controls for each workstation shall be located convenient to the operator’s work stations. Controls shall include means for starting and stopping the machine and other necessary controlling devices. All operations of the machine shall be controlled by an anti-repeat foot switch. When specified (see 6.2), the machine shall be provided with two anti-repeat foot switches to perform two different operations at the same time and independent of each other in accordance with the manufacturer’s standard commercial practice.

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3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910 and ANSI B11.5. if a conflict arises between OSHA 29 CFR 1910 and ANSI B11.5, OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Covers. guards, and other safety devices. Covers, guards, and other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the conmercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration the procuring activity shall provide part design data to the supplier of the machine (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the machine. All guards shall allow access to the guarded parts.

3.2.5.2 Noise level. Unless otherwise specified (see 6.2), noise emitted by the machine when operating under a no load condition shall not exceed 84 DB when measured on the A scale of a type 1 or type 2 sound level inter at slow response.

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During the manufacture, service, transportation, or storage of the machine, and its components, the machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the machine at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all riming parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, filler caps, and drain plugs shall be accessible.

3.2.10 Interchangeability. All parts having the sane manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

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3.3 Construction, The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer’s commrcial practice.

3.3.4 Welding. brazing. or soldering. Welding, brazing, or soldering shall be employed only where specified In the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer’s commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable “Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emissision ‘and susceptibility requirements shall be as specified.

3.4 Co orients. The machine shall consist of not less than a base and franc, hyraulic drive, punch station, structural shape shear station, coping station, bar shear station, and electrical equipment.

3.4.1 Base and franc. The base and franc shall either enclose or support the metalworking stations, hydraulic drive system, and all other components specified herein. The base and frame enclosure shall have adequate strength, rigidity, and shock resistance to withstand all stresses produced when operating the machine at its maximum capacity without permanent deformation or premature failure. The machine shall be provided with a lifting eye or other means for lifting at or near its balance point on tie frame. The eye or lift point shall be capable of supporting five times the weight of the machine, and its location shall not cause distortion to the frame when the machine is suspended.

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3.4.2 Hydraulic drive. Unless otherwise specified, the four metal-working stations shall be powered by one or more hydraulic cylinders. When specified, two hydraulic cylinders shall power the four permanent metalworklng stations (see 6.2). The hydraulic drive shall include a hydraulic pump fluid reservoir, ram cylinder unit, and connecting hose or tubing. The hydraulic drive shall be kept clean by eans of filters which ensure proper operation of the system. The hydraulic drive shall operate on non-flanmable hydraulic fluid.

3.4.2.1 Hydraulic pump. The hydraulic fluid pump shall be of the rotor drive type. The pump motor shall conform to the requirements of 3.4.7.1. The pump shall be provided with hydraulic fluid control devices which shall include a directional control valve and a pressure relief valve. The pressure relief valve shall be adjusted with a relief setting of not more than 10 percent higher than the highest working pressure of the system.

3.4.2.2 Reservoir. The hydraulic fluid reservoir shall have sufficient volumetric capacity to supply the pump volume and circuit operating requirements, and maintain the fluid temperature within its recommended temperature range, while operating the machine at maximum capacity. The reservoir shall be provided with an accessible pump, suction filter, filler strainer, sight level gage, magnetic drain plug and a covered access cleanout opening.

3.4.2.3 Ram cylinder unit. The ram cylinder unit shall consist of a double acting cylinder connected to the frame with the piston connected to the ram The cylinder shall be sealed against leakage of hydraulic fluid.

3.4.2.4 Connecting hose or tubing. Hydraulic fluid transfer between the pump and the ram cylinders shall be accomplished through the connecting hose or tubing. The hose or tubing shall conform to NFPA T2.24.1.

3.4.3 Punch station. The punch station shall consist of a punch holder, die holder, and stripper. The punch station shall be capable of punching holes of the dimensions specified in table I. The station shall have a throat depth in accordance with table I.

3.4.3.1 Punch holder. The machine shall be provided with a punch holder for holding tie punch In punching position under the ram cylinder unit. The punch holders shall be constructed of alloy steel and securely hold the punch in position. The punch holders shall assure alignment with the punches and dies. The punch holders shall be designed to accommodate standard round and shaped punches.

3.4.3.2 Die holder. The machine shall be provided with a die holder for holding the die in position. The die holders shall be constructed of hardened steel and precision ground to securely hold the dies in such a manner to prevent the dies from turning. The die holders shall be designed to accommodate standard round and shaped hole dies.

3.4.3.3 Stripper. The machine shall be provided with a stripper or other means for automatically releasing the workpiece from the punch when the punch is withdrawn from the die.

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TABLE I. Sizes and characteristics.

Characteristics	Size									
	1	2	3	4	5	6	7	8	9	10
Punch Capacity:										
Punch, Nominal tons	35	45	50	70	70	90	90	105	120	150
Throat Depth, inches	5	9	6	9	20	12	20	20	20	24
1/ Strokes Per minute	28	35	35	30	30	40	40	42	40	40
Hole diameter, inches in	13/16	13/16	13/16	1-1/16	1-1/16	1-1/8	1-1/8	1-5/16	1-1/4	1-9/16
Material thickness, inches	3/8	5/8	3/4	3/4	3/4	1	1	1	1-1/8	1-3/16
Coping capacity (square):										
In plate thicknesses, inches	1/4	1/4	1/4	3/8	3/8	5/16	5/16	1/2	3/8	5/16
Depth of cut, inches	2	1-3/4	1-3/4	2-3/8	2-3/8	3	3	3-1/2	2-1/2	3
Width of cut, inches	3	1-3/4	1-5/8	1-3/4	1-3/4	2	2	2	2	3
Bar Shearing Capacity:										
Thickness, inches	5/8	1/2	3/4	3/4	3/4	3/4	3/4	3/4	1	1
Flat bar size, inches	6	4	5	8	8	15	15	10	10	10
Length of blade, inches	13	8	12	18	18	19	19	20	18	18
Structural shapes shearing capacity:										
Round bar, diameter, inches	3/4	1-1/16	1-1/16	1-1/4	1-1/4	1-1/4	1-1/4	1-1/4	1-1/4	1-1/2
Square bar, inches	3/4	1-1/16	1-1/16	1-1/4	1-1/4	1-1/4	1-1/4	1-1/4	1-1/4	1-1/2
Angle, square cut, inches	3x3x5/16	3x3x5/16	3x3x5/16	4x4x1/2	4x4x1/2	6x6x1/2	6x6x1/2	6x6x1/2	6x6x1/2	6x6x5/8
Angle, 45 degree miter, inches (when specified)	2x2x5/16	2x2x5/16	2x2x5/16	3x3x5/16	3x3x5/16	3x3x3/8	4x4x1/2	3x3x5/16	4x4x1/2	4x4x1/2
Drive:										
Motor Horsepower	3	5	5	7-1/2	7 1/2	7-1/2	7-1/2	7-1/2	7-1/2	7-1/2

NOTE: Capacities are based on cross section measurement of 60,000 PSI tensile strength material. Requirements shall be not less than shown.

1/ Strokes per minute are based upon 1/2 inch stroke length and the machine being cycled without a workpiece.

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3.4.3.4 Punches and dies. When required (see 6.1), punches and dies of sizes and shapes spectfied shall be furnished. The punches and dies shall be tool steel, hardened, and ground for accuracy. Each punch and die shall be ground for accuracy. Each punch and die shall be permanently marked for size and identification.

3.4.4 Structural shape shear station. The structural shape shear station shall be capable of shearing structural shapes and bar stock of the dimensions specified in table I. The arrangement of the blades shall be such that structural angle shearing shall be accomplished in a horizontal plane Structural angle shapes shall be sheared at 90 degrees. When specified (see 6.2), structural angle shapes shall be mitered at 15 degrees, 30 degrees, and 45 degrees, both leg up and leg down. The quality of the "drop-off" structural angle cuts shall be such that the two legs of the angle are neither apart by more than 91 degrees, 30 minutes, nor closer together than 88 degrees, 30 minutes. The size and arrangement of the blades shall be in accordance with the manufacturer's standard practice. The blades shall be mounted in such a manner that they can be removed and replaced without the use of special tools. The structural shape shear shall be equipped with an adjustable workpiece stripper or hold-down, capable of accommodating stock of any shape or dimension within the capacity of the machine.

3.4.4.1 Angle iron miter gage. When specified (see 6.2), an angle iron miter gage shall be furnished on the structural shape shear to allow shearing of angle iron at 15 degree, 30 degree, 45 degree, and 90 degree angles without the use of lay-out marks. The miter gage shall be adjustable to accommodate angle iron sizes from 1 in. x 1 in. Up to the angle iron sizes specified in table I. An adjustable stop shall be provided for locating the work in the desired position.

3.4.5 Co in station The coper shall be capable of coping square cuts in metal to meet the requirements of table I. The coper shall be equipped with an adjustable gage-designed to govern the depth of the cut. A guard shall be provided for protecting the operator when the coper is operating but is not being used. The blades of the coper shall be removable and replaceable without the use of special tools.

3.4.6 Flat bar shear station. The bar shear shall consist of an upper moveable blade and a lower stationary blade. The bar shear shall be capable of shearing flat bar stock of the dimensions specified in table I. The bar shear shall be capable of shearing the maximum width capacity specified in table I to a quality that the camber of the sheared edge of the "drop off" cut shall be 1/32 inch or less per 6 inch of bar width. The bar shear shall also be capable of mitering one leg of structural angle to 15 degrees, 30 degrees, and 45 degrees. The shear blades shall be not less than the length specified and shall be mounted in such a manner that they can be removed and replaced. The flat bar shear shall be equipped with a stripper or hold down which shall be adjustable to accommodate stock of any dimensions within the capacity of the machine. The holddown shall be no farther than 4 inches from the shearing plane.

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3.4.7 Electrical requirenmts. The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 3-phase, 60-Hertz (Hz), dual 230/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired as specified (see 6.2).

3.4.7.1 Motors. Motors shall meet the requirements of 3.4.7. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.7.2 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.8 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. 'The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of O to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I for the machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered.

3.6 Physical size limitations. When specified (see 6.2.), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.7 Performance. The machine shall be capable of performing the metalworking operations of punching, structural shape shearing, coping, and bar shearing. The machine shall conform to the requirements of table I. The machine shall be capable of performing metalworking operations on steel workplaces having a minimum tensile strength of 60,000 pounds per square inch (PSI).

3.7.1 Additional performance requirements. Performance features required in addition to, or In lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

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3.9 Optional and accessory equipment. Optional and accessory equipment shall be furnished as specified (see 6.2). The equipment provided shall be functional and compatible to the size machine specified in table I.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units charts, plates, and related 'attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate or chart. When specified (see 6.2), a lubrication plate or chart shall be attached to each machine. If a chart Is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2), the information provided on the chart or plate shall be as listed below:

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.11.2 Namplate. A namplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below:

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE number or military or Federal . specification number)
Type of filter

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

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3.13 Workmanship. Workmanship of the machine and Its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance *of the* inspection requirements specified here in, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 The Inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

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4.6 Acceptance test. An acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility With ambient conditions of 41° to 104° Farenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. Each machine shall be operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the test period.

4.8.3 Performance tests. The machine shall be tested for conformance with 3.7 by performing the following tests without making any adjustments to or performing any maintenance on the machine.

4.8.3.1 Punch test. The machine shall be set up for punching operations. While utilizing the maximum size punch and die set specified in table I, not less than ten holes shall be punched in maximum thickness steel plate of the size specified under punching capacity in table I.

4.8.3.2 Structural shape shearing test. The machine shall be set up for structural shape shearing operations. Not less than one each of the specified shapes of the maximum sizes specified in table I shall be sheared. Angle iron shapes shall be sheared at 90 degrees and mitered at angles of 15 degrees, 30 degrees and 45 degrees both leg up and leg down. The quality of the “drop off” structural angle cuts shall conform with 3.4.4.

4.8.3.3 Coping test. The machine shall be set up for coping operations. Not less than one each of the specified shapes of the maximum sizes under coping capacity in table I shall be coped.

4.8.3.4 Bar shearing test. The machine shall be set up for bar shearing operations. Not less than 6 flat bars cuts of the maximum sizes specified in table I shall be sheared. The quality of the sheared edge of the “drop off” cuts shall conform with 3.4.6.

4.8.4 Additional performance tests. Performance tests required in addition to or In lieu of the requirements of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.5 Noise level test. Noise level measurements shall be made to ensure conformance with 3 2 5 2 Unless otherwise specified (see 6.2), the measurements shall be made at the operator’s work stations and each corner of

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the machine. Noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response. The test shall be conducted with the machine at normal operational speed.

4.8.6 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.8. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.8.7 Electromagnetic interference control test. Equipment requiring electromagnetic Interference control shall be tested for compliance with 3.3.7 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machines covered by this specification are intended for use in punching, shearing structural shapes, coping, and shearing bars of various metals. This machine is intended for use in manufacturer's plants and military and contractor's repair shops.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c* Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify metalworking operations, if different (see 3.2).

f. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

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g. Two anti-repeat foot switches when specified (see 3.2.4).

h. Additional safety and health requirements, if required (see 3.2.5).

i. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5.1).

j. Specify acceptable noise level, if different (see 3.2.5.2).

k. Specify environmental protection requirements exceeding established regulations (see 3.2.8).

l. Specify surface finish requirements, if different (see 3.3.3).

m. Painting, if different (see 3.3.5.).

n. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

o. Two hydraulic cylinders,if required (see 3.4.2).

p. Specify sizes and shapes of punches and dies, when required (see 3.4.3.4).

q. Miter structural angles 15, 30 and 45 degrees in the structural shape shear station, when required (see 3.4.4).

r. Angle iron miter gage, when required (see 3.4.4.1).

s. Voltage requirements, if different (see 3.4.7).

t. Initial voltage, as specified (see 3.4.7).

u. Motors, if different (see 3.4.7.1).

v. Worklight, if required (see 3.4.7.2).

w. Hour meter, if required (see 3.4.8).

x. Size and capacity, if different (see 3.5).

y. Specify physical size limitation, if required (see 3.6).

z. Additional performance features, if required (see 3.7.1).

aa. Specify and fully describe optional equipment, if required. (see 3.9).

bb. Specify applicable repair parts, if required (see 3.10).

cc, Lubrication plate, if required (see 3.11 .1).

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dd . Lubrication plate Information, If different (see 3.11.1).

ee. Nameplate, if different (see 3.11.2).

ff. Hydraulic fluid requirements plate, if required (see 3.11.3).

gg. Hydraulic fluid requirements plate, if different (see 3.11 .3).

hh. Inspection conditions, if different (see 4.3).

ii. First article inspection, if different (see 4.4).

jj. Quality conformance inspection, if different (4.5).

kk. Acceptance test, if different (see 4.6).

ll. Additional performance tests, if required (see 4.8.4).

mm. Specify location of noise level test, if different (see 4.8.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction With the specific acquisition to ensure that only essential data is requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production Item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the

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Government reserves the right to waive the requiremnt for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries ayW differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International d'Unites (S1)," of the International Bureau of Heights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurenmts may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

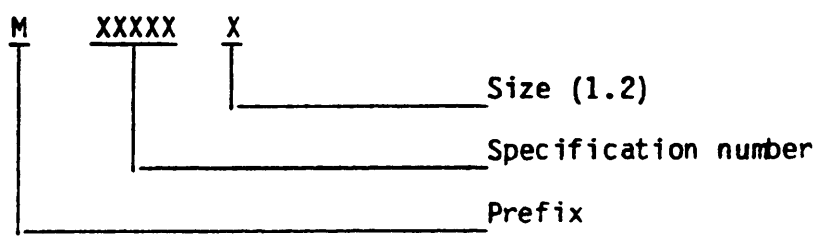
6.7 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.8 Training. Training required by the procuring activity should be provided as specified in the contract.

6.9 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.10 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.11 Part or Identifying Numbers (PINs). The PIN to be used for the machines acquired to this specification are created as follows:



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6.12 Subject term Keyword listing

Combination metalworking machine
Ironworker (hydraulic)
Metalworking machine
Steelworking machine

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Army - AV
Navy - MC

Preparing Activity:
DLA - IP

Project (3445-0126)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-80279	2. DOCUMENT DATE (YYMMDD) 27 Nov 91

3. DOCUMENT TITLE
PUNCHING, SHEARING, NOTCHING, AND COPING MACHINE, METAL, HYDRAULICALLY DRIVEN, 4 STA TYP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) COMMERCIAL (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	b. TELEPHONE (Include Area Code) (1) Commercial: (901) 775-6269 (2) AUTOVON: 683-6269
c. ADDRESS (Include Zip Code) 2163 Airways Blvd Memphis, TN 38114-5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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